E-Learning Platform – SQL Data Analysis Project

Project Overview

Analysed an E-learning platform's purchase data to understand learner behaviour, course performance, and revenue generation using SQL. The objective was to extract meaningful business insights to support data-driven business decisions.

Tools & Technologies

- **Database:** MySQL
- **Language:** SQL
- **Concepts:** Joins, Aggregate Functions, GROUP BY, HAVING, Date Functions, Filtering, and Sorting.

Database Tables

- **Learners:** Learner ID, Full Name, Country.
- **Courses:** Course ID, Course Name, Category, Unit Price.
- **Purchases:** Purchase ID, Learner ID, Course ID, Quantity, Purchase Date.

Key Business Insights

1. **Course-wise Revenue:** Analysed total revenue generated by each course.
2. **Learner Spending:** Identified learners based on their total spending.
3. **Country-wise Revenue:** Evaluated revenue contributions from different countries.
4. **Learner Distribution:** Analysed the number of learners across countries.
5. **Average Purchase Value:** Calculated the average value of purchase transactions.
6. **Monthly Revenue Trends:** Examined monthly revenue patterns to understand sales performance.
7. **Top 3 Purchased Courses:** Identified the three most popular courses based on quantity sold.
8. **Category-wise Analysis:** Compared revenue and unique learners across course categories.
9. **Multi-category Learners:** Identified learners purchasing courses from multiple categories.
10. **Unpurchased Courses:** Identified courses without any purchase records.

Conclusion

The analysis provides insights into revenue performance, popular courses, learner spending patterns, geographical distribution, and course demand. These findings can help businesses improve marketing strategies, optimise course offerings, and identify growth opportunities.